

MORUYA AP, Australia

WMO: 959370

Lat: 35.9003S

Lon: 150.1436E

Elev: 15

StdP: 14.69

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.1	36.1	26.4	20.3	54.7	29.1	23.1	52.8	20.2	57.7	18.0	58.6	2.9	250	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	84.4	66.2	79.6	68.1	77.3	68.4	72.9	77.6	71.7	76.4	70.7	75.2	12.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.4	115.8	74.9	70.2	111.1	74.1	69.0	106.8	73.4	36.6	78.1	35.5	76.5	34.6	75.2	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.3	19.1	16.9	DB	29.1	99.8	1.6	6.4	27.9	104.4	27.0	108.2	26.0	111.8	24.8	116.4	
			WB	28.8	74.8	1.6	2.0	27.7	76.3	26.7	77.4	25.8	78.5	24.7	79.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	68.5	68.3	66.0	61.6	56.2	52.7	50.7	51.8	55.3	58.9	62.3	65.4
	DBStd	7.48	4.19	3.77	3.70	3.71	3.81	3.71	3.87	4.28	4.90	5.32	4.70	4.05
	HDD50	85	0	0	0	0	1	15	36	26	6	1	0	0
	HDD65	2309	11	7	31	112	275	370	443	411	295	204	107	43
	CDD50	3646	572	514	497	349	193	94	58	80	165	277	369	478
	CDD65	396	118	101	63	11	1	0	0	4	16	26	56	56
	CDH74	1816	496	372	214	73	16	0	1	6	84	200	164	190
	CDH80	433	114	63	28	19	1	0	0	17	79	63	49	49
(14) Wind	WSAvg	7.3	8.4	7.6	7.6	6.7	6.2	6.4	6.2	6.9	7.1	8.1	8.4	8.2
(15) Precipitation	PrecAvg	39.9	3.6	3.9	4.6	3.9	3.1	3.5	2.0	2.5	2.6	3.5	3.6	3.2
	PrecMax	69.1	14.3	22.7	22.5	19.9	10.2	20.2	8.8	12.2	7.8	20.7	11.9	9.7
	PrecMin	18.4	0.4	0.1	0.2	0.1	0.0	0.0	0.0	0.1	0.3	0.4	0.4	0.4
	PrecStd	12.6	2.8	4.8	4.3	4.3	2.6	4.0	2.1	2.9	1.9	3.7	2.6	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.4	88.9	83.1	81.7	75.9	69.3	70.9	74.4	82.3	90.3	88.4	86.1
		MCWB	70.6	71.1	67.6	61.8	60.0	56.1	54.1	55.7	58.1	62.6	65.3	67.1
	2%	DB	81.2	79.6	78.0	74.9	71.5	65.9	66.0	69.0	75.4	78.9	77.4	78.1
		MCWB	70.7	71.7	69.5	64.5	59.9	55.5	53.1	54.3	57.2	59.9	65.5	68.7
	5%	DB	78.0	77.6	76.0	72.6	68.6	63.7	63.5	65.2	70.1	72.4	73.1	75.1
		MCWB	70.4	70.6	68.5	64.4	58.7	54.2	52.4	53.3	56.4	60.5	65.8	68.0
	10%	DB	76.1	76.0	74.3	70.7	66.1	61.8	61.2	62.7	66.1	68.8	71.0	73.5
		MCWB	69.4	69.4	67.3	63.1	57.1	53.4	51.4	52.4	56.9	61.1	64.5	67.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.1	74.7	73.5	69.3	64.8	60.9	58.8	59.9	62.7	66.7	70.4	72.0
		MCDB	80.8	79.0	76.0	73.9	69.7	64.5	63.3	66.7	70.2	72.6	75.8	77.5
	2%	WB	72.7	73.0	71.2	67.4	62.2	58.3	56.4	57.6	61.0	64.9	68.2	70.4
		MCDB	78.0	77.3	75.3	72.1	68.0	62.5	62.1	64.1	67.4	70.7	73.7	75.1
	5%	WB	71.6	71.8	70.0	65.8	60.4	56.6	54.4	56.0	59.5	63.5	66.8	69.2
		MCDB	76.8	76.1	74.3	70.6	65.9	60.8	60.3	62.1	66.0	69.6	72.0	73.8
	10%	WB	70.5	70.7	68.7	64.1	58.8	55.0	52.7	54.3	57.9	61.9	65.3	68.0
		MCDB	75.2	74.9	73.2	69.0	64.5	59.4	58.9	60.6	64.9	68.1	70.3	72.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	13.9	15.2	17.8	20.7	18.8	21.7	22.1	21.4	19.0	16.4	14.6
		MCDBR	18.7	17.1	18.6	21.6	24.8	21.7	24.5	25.8	29.4	28.0	22.1	18.5
	5% WB	MCWBR	10.0	9.7	11.2	12.9	14.7	13.1	14.3	14.9	13.9	12.8	12.1	10.5
		MCDWR	15.8	15.0	15.6	18.5	20.8	17.6	21.3	21.7	22.8	20.7	19.0	15.5
(40) Clear-Sky Solar Irradiance	taub	0.423	0.383	0.355	0.329	0.312	0.299	0.290	0.299	0.333	0.351	0.374	0.382	0.382
		taud	2.346	2.503	2.559	2.608	2.600	2.632	2.653	2.620	2.519	2.501	2.473	2.468
	Ebn at Noon	290	295	292	282	269	265	275	288	294	302	303	304	304
		Edn at Noon	42	35	30	25	23	20	21	25	31	34	37	37
(44) All-Sky Solar Radiation	RadAvg	2061	1776	1518	1181	916	709	836	1108	1507	1810	1949	2114	2114
	RadStd	169	147	112	96	46	54	66	58	102	123	149	140	140

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (26 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-135	+225	+94

Nomenclature: See separate page